

Principle of Information security• Section-A ($2 \times 3 = 6$)

1. What is Countermeasures in terms of information security?

Ans Countermeasures refer to the techniques, processes or tools used to reduce security risks, detect attacks and prevent threats to information systems. These measures help protect systems, networks and data from unauthorized access, breaches and damage. Examples include firewalls, encryption, intrusion detection system (IDS), antivirus software and multi-factor authentication (MFA).

2. What are Intruders?

Ans Intruders are unauthorized individuals or programs that attempt to access, modify or damage a computer system or network. Their aim is often to steal information, disrupt services, or install malicious software. Intruders can be classified into:

- Hackers (who exploit vulnerabilities for fun or challenge)
- Crackers (who break into systems for malicious purpose)
- Insider threats (employees or associates misusing access privilege).

3. Difference Betⁿ. Intruders, viruses and Firewalls with examples

Ans

Term	Definition	Example
• Intruder	→ A person or software trying to gain unauthorized access	Hacker trying to access a bank server
• Virus	→ A malicious software that infects files and spreads.	ILOVEYOU virus that spread via email.
• Firewall	• A security system that monitor and controls network traffic.	Windows Defender Firewall

• Section - B (5x8=15)

1. What is virus? Explain all types of viruses.

Ans: A virus is a type of malicious software (malware) that attaches itself to legitimate programs or files and replicates when executed. It can corrupt, delete or steal data, disrupt system operations, and spread to other systems.

• Types of Virus -

(i) File Infector Virus -

- Attaches itself to executable file (.exe, .com).
- Activates when the infected file is run.
- Ex - ~~Chenobyl~~ CMH (Chenobyl) Virus.

(ii) Boot Sector Virus -

- Infects the master boot record (MBR) of hard drive or USBs.
- Activates when the system is booted.
- Ex - Michelangelo virus.

(iii) Macro Virus -

- Written in macro languages (eg - MS Word/Excel).
- Spreads through infected documents.
- ex - Melissa virus.

(iv) Polymorphic Virus -

- Changes its code each time it spreads to avoid detection.
- Hard to detect with signature-based antivirus.
- ex - Storm Worm.

(v) Resident Virus -

- Install itself in system memory and infects files automatically.
- Doesn't need the original infected file to spread.
- ex - Randex, CMJ

2. What is Password Management? Explain issues related to Password management.

Ans Password management refers to the strategies, tools and practices used to create, store, organize, and secure passwords. It ensures that users use strong, unique, and secure passwords to protect systems and data from unauthorized access.

Ex- Password managers (e.g. LastPass, Bitwarden), Password policies in organizations.

• Issues Related to Password Management :-

(i) Weak Passwords -

- Users often choose easy-to-guess passwords (e.g. '123456', 'password').
- These are vulnerable to brute force or dictionary attacks.

(ii) Password Reuse -

- Using the same password across multiple platforms.
- If one site is compromised others become vulnerable too.

(iii) Forgetting Passwords -

- Complex or frequently changed passwords are hard to remember.
- Leads to account lockouts or helpdesk overload.

(iv) Phishing Attacks -

- Users may be tricked into giving away passwords via fake websites or emails.
- Common method to steal credentials.

(v) Poor Storage Practices -

- Storing passwords in plain text or writing them down.
- Easy target for attackers if device is compromised.

3. Explain Active and Passive attacks and its types.

Ans- Active Attacks - Active Attack involves modifying or disrupting data, making them more dangerous. The attacker actively

interferes with communication.

Types of Active Attacks -

- Masquerade - pretending to be an authorized user.
- Replay Attack - Reuses valid data transmissions to gain unauthorized access.
- Modification Attack - Altering transmitted data.
- Denial of Service (DoS) - Make system or network unavailable to users.

(ii) Passive Attacks - passive attacks involve unauthorized monitoring or intercepts data without altering it. The attacker aims to gather information security.

Types:-

- Eavesdropping (Sniffing) - Intercepting communication to capture sensitive data.
- Traffic analysis - Monitoring transmission patterns to gather information.

Section - C (4x7 = 28)

1. Explain working and classification of Intrusion detection System.

Ans An Intrusion Detection System (IDS) is a security tool used to monitor network or system activities for malicious actions or policy violations.

It helps in detecting unauthorized access, attacks or unusual behavior in real time.

• Working of IDS:-

- (i) Monitoring - Continuously monitors network traffic or system logs.
- (ii) Analysis - Compares activity to known attack patterns or anomalies.
- (iii) Detection - Identifies potential threats or intrusions.
- (iv) Alerting - Sends alerts to administrators for further action.

(v) Response - Some IDS can take automated actions (e.g. blocking IPs).

• Classification of IDS:-

(i) Based on Detection Method:-

- Signature-based IDS - Detects known attacks using predefined patterns.
- Anomaly-based IDS - Identifies deviations from normal behavior.
- Hybrid IDS - Combines both methods for better accuracy.

(ii) Based on Deployment -

- Network-based IDS - monitors network traffic.
- Host-based IDS - Monitors activities on a single device.

(iii) Based on Response -

- Passive IDS:- only logs and alerts.
- Active IDS - Block threats automatically.

2. What are the advantages and importance of firewall design system?

Ans Need of a Firewall:-

(i) Network Security - protects internal networks from ~~any~~ unauthorized external access.

(ii) Traffic Control - Filters incoming and outgoing traffic based on security rules.

(iii) Threat Prevention - Blocks malware, hackers and cyberattacks.

(iv) Access Management - Restricts access to sensitive data and services.

(v) Compliance - Helps meet regulatory security standards.

• Importance of Firewall Design -

(i) First Line of Defense - Acts as a barrier between trusted and untrusted networks.

(ii) Prevents Data Breaches - Stops unauthorized data leaks.

- and cyber espionage.
- (iii) Enhance Privacy - Shields user data from malicious tracking and attacks.
 - (iv) Reduce Attack Surface - Minimizes vulnerabilities exposed to the internet.
 - (v) Supports Secure Remote Access - Enables safe VPN and remote work connections.

Q. Explain Malware and types of Malware on the basis of infection.

Ans. Malware (Malicious software) is any harmful program designed to damage, disrupt or gain unauthorized access to computer systems.

• Types of Malware Based on Infection -

- (i) Virus -
 - ↳ Attaches itself to clean files and spreads when the infected file is executed.
 - ex: File infector virus, macro viruses.
- (ii) Worm -
 - ↳ Self-replicating malware that spreads over networks without user interaction.
 - ex: ILOVEYOU worm, Stuxnet.
- (iii) Trojan Horse -
 - ↳ Disguises itself as legitimate software but performs malicious actions when executed.
 - ex: Remote access Trojans (RATs), Banking Trojans.
- (iv) Spyware -
 - ↳ Secretly monitors user activity and sends data to attackers.
 - ex: Keyloggers, Adware with spy functions.

(iv) Ransomware - Encrypts victim's files and demands payment for decryption.

ex: WannaCry, Locky.

(vi) Rootkit - Gains admin-level access to a system and hides malicious processes.

ex: Bootkit, Firmware rootkits.

4. What is Firewall? Explain @ Types of Firewalls in details.

Ans: A Firewall is a network security device or software that monitors and controls incoming and outgoing traffic based on predefined security rules. It acts as a barrier between trusted internal networks and untrusted external networks to prevent unauthorized access, cyber attacks and data breaches.

• Types of Firewalls :-

(i) Packet Filtering Firewall -

• works at: Network layer

• Function - Examines each packet's source/dest IP, Port No, and Protocol.

• Pros - Fast, low resource usage.

• Cons - No deep inspection, vulnerable to IP spoofing.

ex: Basic router firewalls.

(ii) Stateful Inspection Firewall (Dynamic Packet Filtering) -

• works at: - Network & Transport layer.

• Fun - Track active connections and allow only legitimate traffic based on connection state.

• Cons - Slower, consumes more resources.

ex: Cisco ASA, Fortigate.

(iii) Proxy Firewall -

works at: Application layer

Fun - Acts as an intermediary betⁿ user and servers, inspecting entire packets (including HTTP/SMTP).

• Pros - Deep inspection, hide internal n/w.

• Cons - High latency, not suitable for high-speed n/w.

ex: Squid proxy.

(iv) Next Generation Firewall -

works at: - Multiple layers (layer 3-7)

• Fun - Combined deep packet inspection, intrusion prevention and malware detection.

• Pros - Advanced threat protection.

• Cons - Expensive, complex to configure.

(v) Unified Threat Management (UTM) Firewall -

• works at - Multiple layers (All-in-one security)

• Fun - Integrated Firewall, antivirus, VPN, spam filtering and IDS/IPS.

• Pros - Simplified security management.

• Cons - Performance overhead.

ex: Sophos UTM, Fortinet UTM.

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